



A review on empirical studies in explainable artificial intelligence

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Abstract

As artificial intelligence (AI) systems become more integrated into decision-making processes, the need for explainability has emerged to foster trust, understanding, and effective human-AI collaboration. With the variety of explainable AI (XAI) methods available, selecting the right one for a specific user group and a given use case remains challenging, especially given the limited empirical validation of existing theoretical guidance. This systematic literature review addresses this gap by synthesizing human-grounded evaluations of XAI methods to identify the influence of specific explanation properties on user outcomes across diverse settings. Moving beyond high-level taxonomies, we classify XAI methods along multiple dimensions, such as scope and output type. Based on this classification, we analyze how the properties of XAI methods affect different user groups, tasks, and domains. Our findings underscore the necessity of context-aware method selection, as the effectiveness of XAI methods varies significantly across use cases. Moreover, our analysis reveals imbalances in the existing empirical landscape, where certain methods and user groups are overrepresented while others are largely overlooked. By validating theoretical proposals with empirical evidence, this review provides actionable guidance for selecting XAI methods that are aligned with user needs and use case demands, paving the way for more targeted and effective human-AI interaction.

Keywords User-centric XAI · Human-grounded evaluation · Literature review · XAI method selection

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1 Introduction

As artificial intelligence (AI) systems become increasingly prevalent in high-stakes domains such as healthcare, finance, and criminal justice (Kalasampath et al. 2025), the need to explain the decisions and predictions made by these systems has grown significantly (Sahoh and Choksuriwong 2023). Traditional machine learning models, such as decision trees, are comparatively transparent by design. However, modern AI systems, particularly deep learning and complex ensemble methods, often operate as black boxes, making it challenging to comprehend how they arrive at specific results or predictions (Arrieta et al. 2020; Lipton 2018). Explainable AI (XAI) aims to address this issue by developing methods that provide meaningful explanations while maintaining the predictive power of advanced AI systems (Adadi and Berrada 2018; Kalasampath et al. 2025).

With the increasing ubiquity of AI systems, the range of user groups engaging with them is growing more diverse. Different stakeholders require different levels of detail and different types of explanations (Brennen 2020). For instance, a medical professional using an AI-assisted diagnosis tool requires a highly detailed explanation referencing medical evidence, while a patient receiving a diagnosis may prefer an intuitive summary. Similarly, explanations in high-stakes applications, such as autonomous driving or financial decision-making, must be precise and reliable (Sahoh and Choksuriwong 2023), whereas explanations in consumer-focused recommendation systems might prioritize understanding and usability.

The motivations for enhancing the interpretability of AI systems as well as their roots are manifold. Users are more likely to trust and adopt AI recommendations if they can comprehend the reasoning behind them (Langer et al. 2021). Additionally, regulatory frameworks, such as the European Union's General Data Protection Regulation (GDPR) and the emerging AI Act, increasingly require AI systems to justify their decisions, particularly when they impact the rights of individuals (Goodman and Flaxman 2017). Moreover, in decision support systems, explanations empower users to scrutinize the decisions of the AI system by providing insights into its rationale (Kostopoulos et al. 2024). Given these differences, there is no universal approach for generating explanations that suits all users and use cases.

XAI research has increasingly moved toward user-centric XAI methods that tailor explanations to the needs and preferences of specific user groups (Liao et al. 2020). Following this paradigm, theoretical contributions have proposed structured frameworks for selecting XAI methods based on user and task characteristics. These contributions range from conceptual models that formalize relationships between user characteristics and explanation types to practitioner-oriented design toolkits intended to support applied decision-making. For instance, Hashemi et al. (2025) introduced a theoretically grounded mapping function that links user types and their explanation needs to appropriate XAI methods. Similarly, Liao et al. (2020) emphasized the importance of aligning XAI methods with user questions and information goals, offering a practitioner-oriented design toolkit. Although these frameworks offer valuable conceptual guidance, a critical gap remains. There is a lack of systematic empirical evidence validating whether these frameworks consistently lead to measurable user-centered benefits across contexts. Existing reviews of XAI evaluation studies have focused primarily on the methodological approaches used to assess XAI methods, rather than on synthesizing empirical findings across studies (e.g., Al-Ansari et al. (2024); Kim et al. (2024a); Löfström et al. (2022); Nauta et al. (2023); Rong et al. (2023)). Consequently, comprehensive empirical evidence on how specific XAI properties (i.e., properties

of XAI methods such as scope or output type) are perceived and received across diverse application contexts and user groups remains limited.

We aim to bridge this gap by conducting a systematic review of empirical studies evaluating XAI methods. Our analysis investigates which XAI properties (e.g., local vs. global) effectively support key user outcomes (e.g., trust, understanding, and usability) across different tasks, domains, and user groups. To this end, we systematically collect and categorize empirical XAI user studies, analyzing their methodological settings and the reported effects of XAI methods. Synthesizing these results provides an evidence-based overview of the current landscape, identifying which types of XAI methods work best in which contexts. Additionally, we reveal understudied areas in the empirical evaluation of XAI methods, thereby offering concrete directions for future research.

The remainder of this paper is structured as follows. Section 2 deals with existing XAI taxonomies, theoretical frameworks guiding the selection of appropriate XAI methods, and evaluation methodologies in XAI research. Moreover, it discusses the research gap addressed by our work. Section 3 describes the methodology we applied in this systematic review, followed by a detailed description of the analytical process applied to the identified empirical studies in Sect. 4. Section 5 presents the results, providing an overview of the analyzed studies as well as quantitative and qualitative analyses. Section 6 discusses theoretical and practical implications, limitations, and directions for future work. Finally, Sect. 7 concludes the paper with the major findings provided by the systematic review and an outlook on future work.

2 Background

2.1 Foundations for guiding the selection of XAI methods

The XAI landscape is rapidly evolving, with a large number of studies proposing novel XAI methods (Kalasampath et al. 2025). While the rapid growth and diversification of XAI methods enhances technical capabilities, it also introduces challenges in identifying the most suitable XAI method for a given application context and specific user requirements (Liao et al. 2020). The variety of XAI methods, each beneficial for different use cases, makes it difficult to establish generalizable guidance on which XAI method should be used in which context. Developing such guidance requires a structured analysis of the research field, which, in turn, depends on a meaningful classification of XAI methods according to their characteristics (i.e., the XAI properties) (Arya et al. 2019). XAI taxonomies provide the foundation for this classification by defining consistent dimensions along which XAI methods can be compared, enabling researchers and practitioners to identify the most suitable XAI properties for different user groups, tasks, and domains (Arrieta et al. 2020; Vilone and Longo 2021). While some taxonomies include implicit recommendations or discuss contextual trade-offs, they generally do not offer direct selection guidance. Rather, they form the essential theoretical basis for deriving it.

Numerous taxonomies have been proposed in the XAI research field to enable the classification of XAI methods (e.g., Angelov et al. (2021); Arrieta et al. (2020); Arya et al. (2019); Belle and Papantonis (2021); Vilone and Longo (2021)). Speith (2022) as well as Schwalbe and Finzel (2024) have reviewed these proposals and developed more comprehensive tax-

onomies. These reviews offer comprehensive categorization schemes that facilitate understanding XAI methods based on distinct characteristics.

Speith (2022) introduces a taxonomy that classifies XAI methods based on six core dimensions: *stage*, *applicability*, *scope*, *result*, *functioning*, and *output type*. The stage dimension differentiates between *ante-hoc* XAI methods (i.e., models that are inherently interpretable, such as decision trees or linear models) and *post-hoc* ones (i.e., explanations generated after model training). The applicability dimension distinguishes between *model-agnostic* XAI methods, which can be applied to any type of model, and *model-specific* XAI methods, which are tailored for particular models (e.g., deep neural networks). The scope of an XAI method determines whether it provides *local* explanations (i.e., explanations for individual predictions) or *global* ones (i.e., explanations describing the overall behavior of an AI system). The result dimension categorizes XAI methods by their general output form (e.g., *feature importances*, *examples*), and the functioning dimension focuses on the way an XAI method generates an explanation (e.g., *perturbations*, *example search*). Finally, the output type refers to the presentation of the result (e.g., *visual*, *textual*, or *numerical*).

Schwalbe and Finzel (2024) extend these classifications by introducing dimensions that capture additional methodological and usability aspects. The taxonomy refines the categorization of XAI methods by introducing the dimensions *required input*, *object of explanation*, *level of abstraction*, and *degree of interactivity*. The required input dimension characterizes the information needed by an XAI method to generate explanations, such as *access to model internals* or *input–output pairs*. The object of explanation dimension categorizes XAI methods according to the focus of the generated explanations (e.g., *underlying data*, *inner representations of the AI system*). The level of abstraction dimension addresses whether explanations are provided in *high-level* concepts (e.g., human-understandable reasoning) or *low-level* technical details (e.g., gradients or feature importance scores). Finally, the degree of interactivity determines whether explanations are *static* (i.e., fixed outputs) or *interactive*, allowing users to dynamically explore different aspects of the model's reasoning. While the interactivity dimension reflects developments in XAI research toward more dynamic and user-engaged explanation interfaces, the remaining additional dimensions primarily refine the characterization of XAI methods at a more granular level.

Although taxonomies help structure the XAI landscape (Schwalbe and Finzel 2024), they do not offer concrete guidance on selecting the most appropriate XAI method for a given use case. Classifying XAI methods based on the XAI properties provides a foundational understanding of their characteristics (Schwalbe and Finzel 2024). However, identifying which methods support human-AI collaboration across different use cases best requires further analysis (Langer et al. 2021). To address this gap, several studies provide guidance for selecting appropriate XAI methods in the context of specific use cases and user needs (e.g., Hashemi et al. (2025); Liao et al. (2020)). These studies emphasize that the performance of XAI methods is closely tied to the context in which they are applied and the needs of the users involved.

Hashemi et al. (2025) recognize the lack of guidance for matching XAI methods to specific user needs. The study analyzes XAI properties and considers different user groups and their needs (e.g., trust, transparency, regulatory compliance) as well as the favored explanations. The study proposes a mapping function that links specific user needs to appropriate XAI methods, thereby facilitating the selection of the best XAI method for a given need. A key limitation of the study is its lack of empirical validation, i.e., the provided map-

ping function that aligns user needs with appropriate XAI methods lacks validation through human-grounded evaluations. Moreover, the proposed alignment is largely derived through theoretical reasoning and expert-informed assumptions. As a result, the extent to which the suggested mappings generalize across diverse user groups, tasks, and domains remains unclear.

Liao et al. (2020) identify links between XAI methods and user needs through interviews with 20 software design experts. The study presents an "algorithm-informed XAI question bank" (Liao et al. 2020), i.e., a list of questions that represent typical user needs for understanding the underlying AI system (e.g., "why/how is this instance given this prediction?"). The study provides valuable insights into designing XAI systems and proposes suitable XAI methods for a specific set of questions, i.e., user needs. The study also uncovers different motivations for seeking explanations, such as gaining further insights, building trust, debugging errors, or satisfying curiosity. As a limitation, the study relies on interviews with software design experts rather than directly interacting with end-users. While these experts offer valuable insights into design considerations, their perspectives do not fully represent the actual needs and perceptions of the end-users, who interact with XAI methods in real-world contexts.

Both Liao et al. (2020) and Hashemi et al. (2025) offer valuable contributions to guide the selection of XAI methods. Nevertheless, the proposed guidance remains unvalidated through human-grounded evaluations. To address this gap, a more in-depth analysis of empirical evaluation findings is required (Hashemi et al. 2025; Liao et al. 2020).

2.2 Evaluating XAI

Doshi-Velez and Kim (2017) recommend assessing XAI methods through a combination of *functionally-grounded and human-grounded evaluations*. A functionally-grounded evaluation (e.g., Elkhawaga et al. (2024)) assesses explanations based on technical proxy measures such as *fidelity* and *stability* (Agarwal et al. 2022). These measures provide an essential foundation for ensuring that an explanation accurately represents the decision-making process of an AI system (Agarwal et al. 2022). However, such evaluations alone do not guarantee that explanations are useful to humans (Doshi-Velez and Kim 2017), as the perception of an explanation depends on cognitive and psychological factors as well (Miller 2019). Studies have shown that, even if technically correct, explanations can be ineffective if they do not align with human expectations or cognitive constraints (Rieffle et al. 2022). Given that users bring varying levels of domain expertise, prior knowledge, and cognitive biases to the interpretation process, explanations must be tailored to their individual needs to be effective (Kim et al. 2024b).

To address this issue, human-grounded evaluations have gained increasing attention. These approaches evaluate XAI methods in controlled environments, typically through user studies in which participants interact with different explanations (Doshi-Velez and Kim 2017). The metrics used to evaluate XAI methods, often referred to as *constructs*, can be categorized as either *implicit* or *self-reported*. Self-reported constructs (e.g., *trust*, *satisfaction*) are derived from user feedback and reflect users' perceptions, whereas implicit constructs (e.g., *task performance*, *task completion time*) are automatically measured while users engage in specific tasks (Naveed et al. 2024).

While functionally-grounded evaluations ensure technical validity, human-grounded ones are essential for assessing whether or not explanations actually support users in practice (Doshi-Velez and Kim 2017). Given our objective to derive empirically grounded guidance for selecting XAI methods based on contextual factors, this review focuses exclusively on human-grounded evaluation studies.

As interest in human-grounded evaluation is growing, an increasing number of studies are evaluating different XAI methods for various use cases with diverse user groups. These studies provide valuable insights into the performance of XAI methods in real-world settings. Human-grounded evaluations are conducted in diverse ways, varying in experimental design, explanation types, user populations, and measured constructs. Recent systematic literature reviews provide structured overviews of human-grounded XAI evaluation. Al-Ansari et al. (2024) systematically analyze 101 user studies to characterize how XAI systems are evaluated through user-centered approaches in HCI. They examine the prominent XAI goals pursued across ten domains, the AI systems and XAI methods adopted, and the methodologies applied in user-centered evaluations. Their analysis concludes in a six-pillar framework that researchers can follow when designing user-centered XAI evaluations. Kim et al. (2024a) review 73 papers describing 77 user studies on AI systems trained with tabular or textual data and propose a taxonomy of human-grounded XAI evaluation. The taxonomy organizes 30 components of meaningful explanations along three dimensions: the contextualized quality of the explanation, its contribution to human-AI interaction, and its contribution to human-AI performance. Both reviews characterize how user-centered XAI evaluation is conducted but do not synthesize which XAI properties produce which effects on user-centric outcomes. Indeed, Kim et al. (2024a) explicitly identify the development of context-sensitive recommendations on XAI method selection as future work.

Beyond synthesizing evaluation methodology, the review by Rong et al. (2023) analyzes user studies to understand how different XAI methods impact various constructs. This review systematically extracts data from user studies, capturing information on the types of explanations used, the measured outcomes, and the evaluation methods employed. The analysis identifies XAI methods that positively influence constructs such as user trust, understanding, satisfaction, and task performance. However, the review by Rong et al. (2023) is subject to several limitations, which prevent it from enabling XAI method selection based on user groups and use case characteristics.

1. Its analysis focuses on the effects of XAI methods on user behavior and experience, rather than on the way specific use case characteristics influence the suitability of different XAI methods. Although the review acknowledges the importance of user diversity, it does not explore how individual differences might affect the effectiveness of different XAI methods. For example, Maxwell and Dumas (2023) emphasize the role of a user-centric design (UCD) methodology in XAI, highlighting that meaningfulness is relative and varies significantly among users and tasks.
2. The review does not systematically analyze the impact of specific XAI properties (e.g., interactivity, scope, output type) on user outcomes. This limits the ability to derive generalizable guidance for selecting XAI methods tailored to the specific needs of different user groups. While the review categorizes studies based on high-level explanation types (e.g., feature importance, rule-based), a deeper examination of the XAI properties within these categories is needed. For instance, Hashemi et al. (2025) explore the

nuanced landscape of user preferences in XAI, considering user characteristics such as domain expertise.

2.3 Research gap

Previous contributions have advanced the discourse on user-centric XAI by proposing conceptual frameworks, practitioner-oriented guidance, and reviews of evaluation practices. However, corresponding efforts remain either largely conceptual or operate at a high level of abstraction. As a consequence, the research field lacks empirical insights on the contextual effects of XAI properties. This gap prevents the validation of existing theoretically grounded guidance (i.e., Hashemi et al. (2025); Liao et al. (2020)) and the development of empirically grounded guidance on selecting XAI methods based on contextual factors. Specifically, it remains unclear how XAI properties (e.g., scope and output type) affect outcomes for different user groups across diverse use cases. Fundamentally, the field lacks a comprehensive assessment of the empirical XAI landscape, which would highlight the XAI properties and use cases that have been extensively examined as well as those that have remained underexplored. Without considering the distribution of evaluated user groups, domains, and XAI properties, interpreting the effectiveness of XAI methods can lead to biased or non-generalizable conclusions (Domingos 2012; Berk 2004).

To address these gaps, our study pursues two interrelated goals.

1. We provide a structured overview of the current state of empirical user studies in XAI. By categorizing studies according to XAI properties, user groups, use case contexts, and evaluation metrics, we identify which XAI properties and use cases have been extensively studied and which remain underexplored.
2. We analyze study results to examine how different XAI properties perform in various settings. This allows us to derive evidence-based guidance for selecting XAI methods tailored to specific user groups and use cases.

3 Methodology

To address the presented research gap, we conducted a systematic literature search across multiple academic databases, focusing on empirical XAI studies. Following the methodology proposed by Kitchenham (2004), we first elaborate on the research questions (RQs), then describe the search strategy as well as the inclusion and exclusion criteria, and finally outline the data collection and analysis process.

3.1 Research questions

This review has two goals:

1. We aim to assess the current state of empirical research on XAI by identifying XAI properties and use cases that have been studied as well as remaining gaps.

2. We aim to derive evidence-based insights into how different XAI properties affect user outcomes in varying use cases.

To achieve these objectives, we must first understand the current landscape of XAI research, i.e., which user groups, tasks, domains, and XAI properties have been the focus of empirical research to date. This will help us to identify potentially imbalanced distributions in the research field and to avoid potential bias in the subsequent analysis. To derive evidence-based insights guiding the selection of XAI methods, we must analyze the effects of XAI properties on user outcomes and the way these effects vary across different use cases. Based on this reasoning, we formulate the following research questions.

RQ1: What is the current state of empirical evaluation in user-centric XAI research?

- **RQ1.1:** Which user groups (e.g., lay users, domain experts, and AI experts), task types (e.g., image processing and decision support), and application domains (e.g., healthcare and finance) have been studied and which remain underrepresented?

- **RQ1.2:** Which XAI properties (e.g., scope, output type, and functioning) have been empirically evaluated and which remain underrepresented?

RQ2: How do specific XAI properties influence user outcomes across different contexts?

- **RQ2.1:** What are the observed effects of different XAI properties on user-centric constructs (e.g., trust, understanding, usability, and task performance)?

- **RQ2.2:** How do these effects vary across different user groups and application domains?

- **RQ2.3:** What actionable insights can be derived to guide the context-sensitive selection of XAI methods?

Our overall aim is to validate prior work in the research field and to develop empirically grounded guidance that supports the selection of XAI methods based on XAI properties, user group characteristics, and use case contexts by systematically answering these research questions. This evidence-based guidance aims to support the design of more effective, user-aligned AI systems that facilitate trust, understanding, usability, and human-AI performance.

3.2 Search strategy

Given the interdisciplinary nature of XAI, we selected five major databases with extensive coverage of AI, human-computer interaction, and behavioral sciences: ScienceDirect, ACM Digital Library, PubMed, IEEE Xplore, and SpringerLink. We focused on publisher-maintained digital libraries to ensure high-quality peer-reviewed sources and precise filtering by metadata fields (title, abstract, keywords). Broader indexing services such as Scopus or Web of Science were not used as primary search engines as their coverage substantially overlaps with the selected databases and may introduce duplicate records or grey literature beyond the scope of this review. To mitigate potential coverage bias, we additionally applied backward snowballing. To ensure that only relevant studies were included, we formulated a search query combining terms related to XAI with terms associated with human-grounded evaluation. Given that empirical user studies in XAI are conducted across multiple disciplines, such as human-computer interaction, psychology, and applied behavioral research, we incorporated both specific evaluation terms (e.g., "user study", "human evaluation") and broader interdisciplinary descriptors (e.g., "behavioral science"). This approach aimed to reduce the risk of omitting relevant human-grounded studies that might not explicitly label themselves as XAI user studies. The resulting query is as follows.

("XAI" OR "explainable AI" OR "explainable artificial intelligence")

AND

("empirical study" OR "empirical research" OR "empirical evaluation" OR "user study" OR "user evaluation" OR "experimental study" OR "human evaluation" OR "case study" OR "behavioral science" OR "behavioural science" OR "user experiment" OR "online experiment" OR "questionnaire" OR "human-grounded evaluation")

To ensure a rigorously structured and reproducible search, this query was applied to the title, abstract, and keywords fields in each database. Conducted in January 2025, the search yielded 838 potentially relevant articles.

To ensure that the included studies contribute meaningfully to the research objectives, we applied a set of rigorous inclusion and exclusion criteria (cf. Table 1). In the following, we discuss and explain the use of these criteria.

1. Since the review focuses on the effectiveness of XAI methods for end-users, studies lacking an empirical, human-grounded evaluation would not provide the necessary information, and, thus, they were excluded.
2. To ensure that all findings are validated through rigorous academic scrutiny, only studies published in peer-reviewed conference proceedings or journals were included. Peer review acts as a quality control mechanism, filtering out studies with weak methodologies or inconclusive evidence (Bence and Oppenheim 2004).
3. Studies were only included if they reported statistically significant results. Given the aim of this review, i.e., to generate actionable insights on selecting XAI methods, studies with inconclusive or insufficient empirical findings would not meaningfully contribute to refining existing theoretical frameworks. Therefore, only studies demonstrating statistically significant findings (i.e., results with a p -value < 0.05) were considered.
4. We required that the XAI properties were transferable to a model-agnostic concept. While model-specific XAI methods are tightly coupled to specific AI systems, model-agnostic methods can be applied across a wide range of AI systems (Devireddy 2025). Therefore, we limited the review to XAI methods whose properties could be conceptualized independently of a specific model architecture to ensure transferability of the synthesized findings.
5. To maintain accessibility and reproducibility, only studies written in English were included. Since most AI research has been published in English, restricting the review to English-language studies ensures consistency and comparability.
6. Studies without accessible full text were excluded as a thorough evaluation of methodologies and findings is essential to assess their contribution to RQ1 and RQ2.

Table 1 Inclusion criteria

Criteria	Include
1. Study evaluates XAI methods with human participants	Yes
2. Study is published in peer-reviewed conference proceedings or journals	Yes
3. Study contains significant results	Yes
4. Properties of the analyzed XAI method are transferable to a model-agnostic concept	Yes
5. Study is written in English	Yes
6. Full text of the study is accessible to the authors of this paper	Yes
7. A more detailed version of the study is included in this review	No

7. If a more detailed version of a study was available, only the most comprehensive version was included to avoid redundancy and to ensure that the review presents the most complete analysis of each contribution.

Adhering to these criteria ensures that the included studies provide robust, human-grounded, and transferable insights that contribute to bridging the identified research gap.

3.3 Data collection

Figure 1 depicts the literature search process. Data extraction and categorization were primarily conducted by one researcher to ensure consistency across studies. To enhance reliability, two additional researchers performed random checks on subsets of the included studies, reviewing both the extracted data and the categorization of XAI properties and constructs. In cases where ambiguities arose, these were resolved through discussion until reaching consensus. This procedure aimed to reduce subjective bias, particularly given the interpretative nature of certain classification decisions (e.g., transferability of XAI proper-

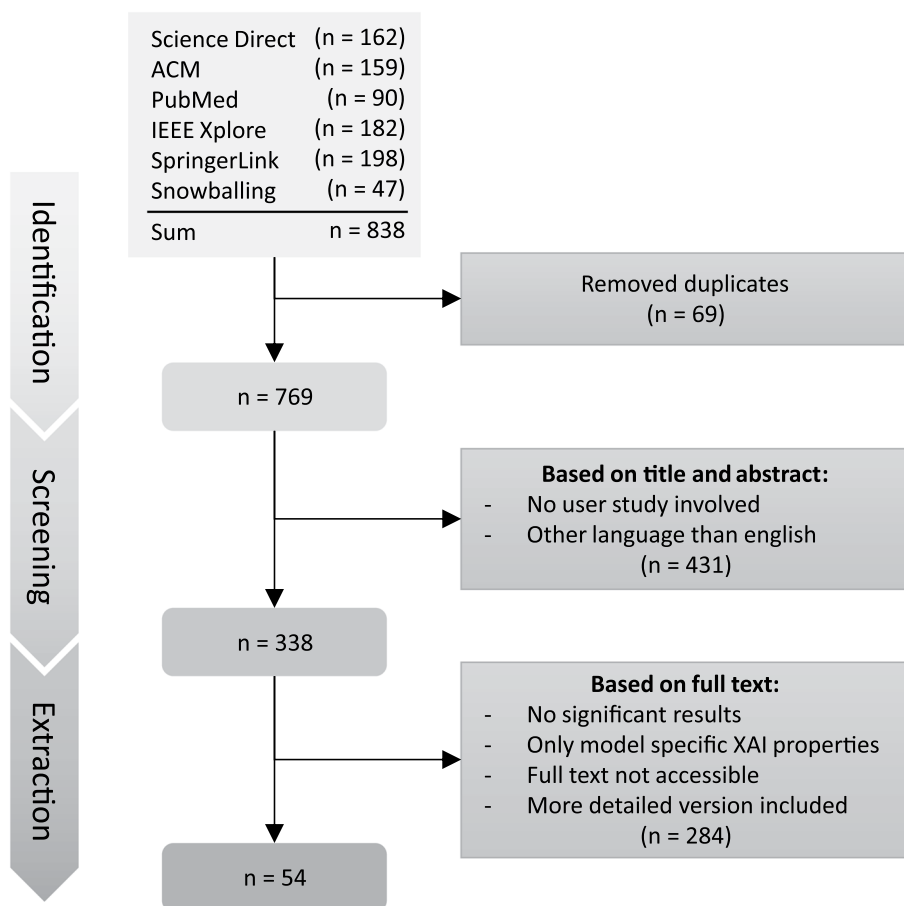


Fig. 1 Flow chart of the literature search process

ties across contexts). The initial search yielded a total of 838 studies, distributed as follows: 162 from ScienceDirect; 159 from the ACM Digital Library; 90 from PubMed; 182 from IEEE Xplore; 198 from SpringerLink; and 47 studies through snowballing, as described by Jalali and Wohlin (2012) and Wewerka and Reichert (2023). After the initial retrieval, the data underwent a multi-step filtering process to refine the selection. First, duplicate records were removed, reducing the total number of studies from 838 to 769. The remaining studies then underwent manual screening of their abstracts to assess whether or not they were aligned with the scope of this review (cf. Table 1, Criterion 1). Studies were excluded if they lacked any form of human-grounded evaluation or if they were written in a language other than English. This step resulted in the removal of 431 studies, leaving 338 for full-text screening.

During the analysis stage, we carefully examined the full texts of the remaining studies to determine if they met all inclusion criteria (cf. Table 1). Studies that did not meet the latter were excluded, resulting in the removal of 284 studies. This rigorous selection process yielded 54 studies for inclusion. Consequently, 54 studies form the basis of this review, offering insights into which XAI methods are most effective for different use cases, and how empirical findings contribute to the broader goal of designing user-centric XAI methods. The following sections of this review discuss the methodology we applied for extracting and synthesizing key findings from these studies.

The included studies were carefully analyzed to extract relevant details and insights. To answer the defined RQs, three types of information were extracted.

1. We extracted information about the XAI methods and, more precisely, the differing XAI properties under observation. This includes attributes regarding the scope, functioning, result, output type, and special XAI properties such as interactivity or more detailed information about the XAI methods.
2. We collected data on the user needs (i.e., the observed user group), goals (i.e., the measured constructs), and further context (i.e., the given task and domain).
3. We extracted every statistically significant finding regarding the measured constructs and conditions.

The extracted information was structured in tabular form, which is included in Online Resource 1.

4 Analysis of XAI user studies

4.1 Categorizing XAI methods

To enable a deeper analysis of the empirical findings, we categorized the different XAI properties according to established taxonomies in the field (cf. Sect. 2.1). Following the taxonomy proposed by Speith (2022), we categorized the XAI methods based on their scope, result, functioning, and output type. We selected this taxonomy as the primary structural framework as it provides a concise and widely adopted categorization of explanation properties that aligns closely with our focus on transferable, model-agnostic XAI methods. While more recent extensions (e.g., Schwalbe and Finzel (2024)) introduce additional dimensions,

several of them either overlap conceptually (e.g., level of abstraction with result and scope) or primarily refine characteristics not central to our analysis. The categories and the corresponding XAI properties are listed in Fig. 2. Additionally, we record special XAI properties highlighted in the studies, such as interactivity and more detailed information on the XAI methods, to account for necessary extensions to the utilized taxonomy.

4.1.1 Scope

The scope of an XAI method refers to its applicability (Speith 2022). *Local* explanations focus on individual predictions, whereas *global* explanations aim to describe the overall behavior of an AI system (Speith 2022).

4.1.2 Result

The result category encompasses the various types of information generated by the XAI method, independent of how this information is generated (cf. Sect. 4.1.3) and presented to the user (cf. Sect. 4.1.4) (Speith 2022). This category is further subdivided into the following XAI properties: *examples*, *counterfactuals*, *feature importances*, *concepts*, *model statistics*, *data statistics*, and *surrogate models*. Examples refer to data points that share similar predictions and characteristics with the input. Counterfactuals illustrate how the input needs to be modified to alter the prediction through alternative data points. Feature importances highlight the impact of input variables on the output. Concepts explain predictions in terms of higher-level abstractions. Model statistics provides specific metrics about the model, whereas data statistics focuses on characteristics of the dataset. Finally, surrogate models approximate the behavior of black-box models using more interpretable representations.

4.1.3 Functioning

The functioning category describes the mechanisms that generate explanations (Speith 2022). This category is further divided into the XAI properties *model evaluation*, *perturbation*, *example search*, *model modification*, and *leveraging structure*. Model evaluation methods produce information by assessing the performance of an AI system (e.g., accuracy). Perturbation methods analyze changes in outputs when modifying inputs. Example search methods generate results by finding data points in the training data. Model modification alters the underlying model to enhance interpretability. Leveraging structure exploits

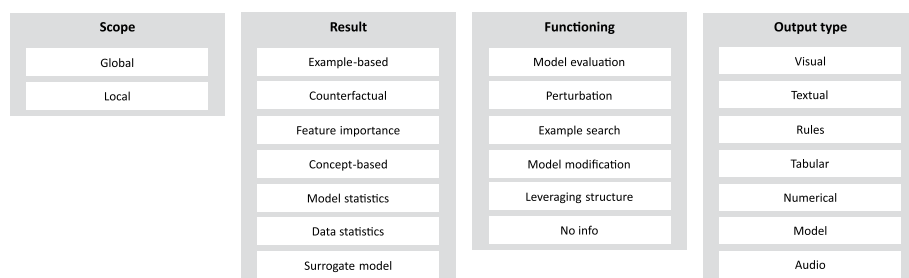


Fig. 2 Categories of XAI properties

the inherent structural properties of the data or model. Finally, we added a "*no information*" label for studies that do not further describe the exact functioning used to produce the explanation.

4.1.4 Output type

The output type category describes the presentation format of the explanation (Speith 2022). Based on the output types prevalent in the analyzed studies, we differentiate between *visual*, *textual*, *rules*, *tabular*, *numerical*, *model*, and *audio* outputs. Visual outputs use graphical representations, such as graphs and heat maps, while textual outputs use natural language descriptions. Rule-based outputs present the decision logic in an if-then format. Tabular formats display the information in a structured way. Numerical outputs provide quantitative indicators, such as scores or weights. Finally, model-based outputs present simplified or interpretable models as explanations, and audio outputs convey the explanations through spoken or auditory cues.

4.1.5 Special properties

We define special properties as unique attributes that are not covered by other categories, but are nevertheless crucial for describing specific XAI methods. These include aspects like *interactivity*, which allows users to engage dynamically with explanations (Bertrand et al. 2023).

4.2 Constructs for evaluating XAI methods

Across the 54 studies we analyzed, 74 constructs were identified that have been used to evaluate XAI methods in user studies. To achieve a more thorough analysis of the study results, we clustered the constructs. To this end, we examined previous studies on human evaluation metrics for XAI methods. We applied the categorization of Rong et al. (2023) who identified four main clusters: *trust*, *understanding*, *usability*, and *human-AI performance*. This categorization was based on the prominent work of Chromik and Schuessler (2020), Hoffman et al. (2018), Lai et al. (2021), and Mohseni et al. (2021). The four primary clusters are further subdivided into nine distinct categories (Rong et al. 2023) (cf. Fig. 3), which differentiate various facets of explainability assessment and facilitate a comparative analysis of XAI method performance in empirical studies. In the following subsections we describe the four clusters, the nine final categories, and the mapping to the underlying constructs (also available in Online Resource 1).

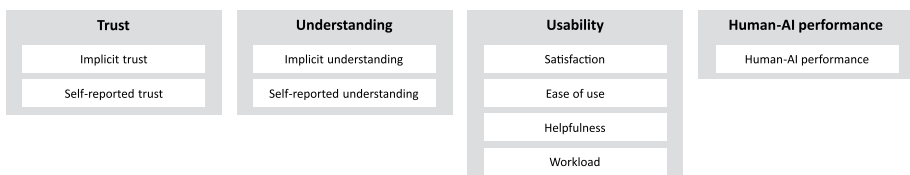


Fig. 3 Construct clusters and the respective subcategories

4.2.1 Trust

In the XAI research field, the notion of trust is essential, as it reflects the users' confidence in the decisions of the AI system as well as their willingness to trust them (Hoffman et al. 2018). Trust measures are further subdivided into *self-reported* and *implicit* measures (Rong et al. 2023). Self-reported measures are typically evaluated with questionnaires that ask about user confidence in following the decision of a system, trust in the system (*global trust*), and trust in a specific decision of the system (*local trust*). This category includes the constructs *agreement*, *confidence*, *deception*, *wariness*, *convincingness*, *reliance*, *global trust*, and *local trust*. Implicit measures focus on the user's willingness to follow the AI systems' decisions and are based on user behavior. The constructs *weight of advice*, *trust calibration*, *persuasive power*, *implicit trust*, and *appropriate trust* are classified within the implicit trust category.

4.2.2 Understanding

Understanding the reasoning and decision-making processes of an AI system constitutes a crucial aspect of XAI, as it allows users to build appropriate mental models of the AI systems (Hoffman et al. 2018). Rong et al. (2023) further subdivide this cluster into *self-reported* and *implicit* measures. Self-reported understanding captures the perceived comprehensibility of the explanations. The constructs assigned to this category include *consent*, *help to comprehend*, *prediction confidence*, *system causability scale*, *transparency*, *detail understanding*, *system understanding*, *subjective learning*, *subjective understanding*, *comprehensibility*, *observability*, and *completeness*. Implicit measures assess how accurately users can demonstrate their understanding of an AI system based on automatically recorded performance during specific tasks. This category comprises the constructs *predictability score*, *prediction accuracy*, *intuitiveness*, *task reflection*, *relative difference*, *recall reconstruction score*, *objective transparency*, *learning effect*, and *objective understanding*.

4.2.3 Usability

The usability of XAI methods refers to how effectively and efficiently users can interact with explanations to support their decision-making processes (Rong et al. 2023). The usability cluster is further subdivided into the following categories: *satisfaction*, *ease of use*, *helpfulness*, and *workload*. Satisfaction captures the users' overall perception of the quality of explanations. It is measured through the constructs *frustration*, *explanation satisfaction scale*, *anger*, *relaxation*, *attitude*, *satisfaction*, *overall evaluation*, *perceived goodness of explanations*, *overall perception*, *perceived explanation quality*, *perceived justification quality*, *information provision*, *informativeness*, and *information sufficiency*. Ease of use assesses the accessibility of explanations and the users' intentions to interact with them. Constructs belonging to this category include *influence performance*, *intention to use*, *controllability*, *usability*, *ease of use*, *global usefulness*, *local usefulness*, and *actionability*. Helpfulness reflects the perceived support that explanations provide to users when interacting with the AI system. It is evaluated through the constructs *perceived task performance*, *competence*, *technical competence*, *helpfulness*, *help to sharpen users' views*, and *self-reported task performance*. Workload measures the mental effort required to process

explanations, which is assessed via *effort*, *cognitive load*, *self-efficacy*, *effectiveness*, and *subjective mental demand*.

4.2.4 Human-AI performance

Human-AI performance evaluation examines the way explanations affect users' efficiency and accuracy when completing tasks (Lai et al. 2021). The constructs associated with this category include *reading time*, *completion time on correct responses*, *task completion time*, *task performance*, and *task completion*.

4.3 Analysis framework

To answer the RQs formulated in Sect. 3.1, we followed a structured analysis framework. This framework combines systematic information extraction, quantitative and qualitative analyses, and their synthesis to provide empirically grounded guidance on selecting XAI methods.

The framework is illustrated in Fig. 4 and involves the following steps.

- We began with a systematic literature search (cf. Section 3), which resulted in 54 user studies.
- We extracted information on the task, domain, and user group involved for each user study. This allowed us to generate an overview of the empirical landscape in user-centric XAI research and to identify the distribution of studied settings. By analyzing this overview, we addressed RQ1.1, i.e., we identified well-studied and underrepresented user groups, tasks, and domains.
- We extracted detailed information about the respective XAI methods that were evaluated in each study. We categorized these according to the scheme described in Sect. 4.1, which encompasses five key property dimensions: scope, result, functioning, output

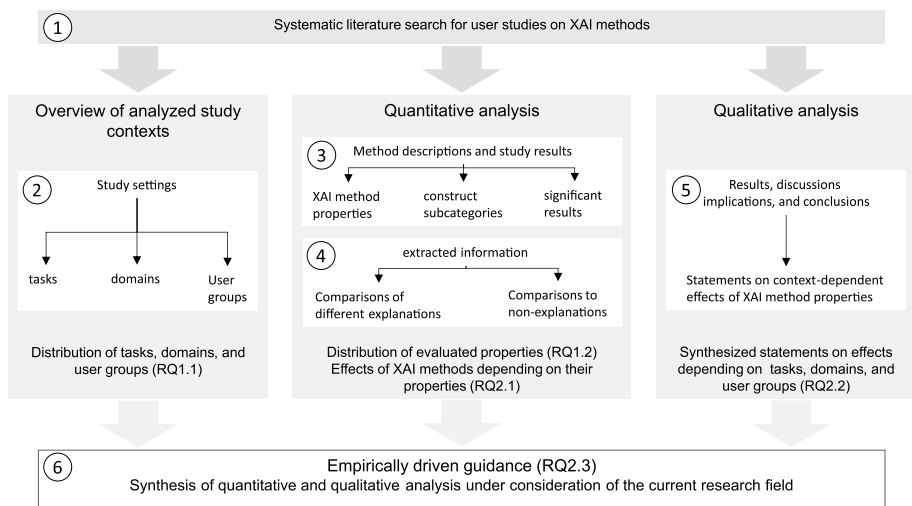


Fig. 4 Flow chart of the analysis framework

type, and special properties (e.g., interactivity). In parallel, we recorded the evaluation constructs used to assess the XAI methods. We mapped all constructs to one of four overarching clusters: trust, understanding, usability, and human-AI performance. Then, we assigned them to one of nine established subcategories (cf. Section 4.2). We extracted every reported statistically significant result and linked it to the relevant construct and the XAI property. This information on XAI properties and study outcomes forms the foundation for our quantitative analysis.

- In this quantitative analysis, we classified studies according to their experimental design. We distinguished between studies that compared explanations to non-explanations (i.e., treatment groups that received explanations compared to control groups that received no explanation) and those that compared different explanations. This distinction was essential as it influenced the way XAI properties were extracted and interpreted. For studies comparing explanations with non-explanations, we extracted all relevant XAI properties, including scope, result, functioning, output type, and any special properties. For studies comparing different explanations, we considered only those XAI properties that varied between the compared explanations. This allowed us to isolate the explanatory factors most likely responsible for the observed effects. Each significant result in these comparative studies was logged as both a positive outcome for the property showing higher performance and a negative outcome for the lower-performing one. In cases in which a method could be assigned to more than one property (e.g., hybrid approaches), we annotated multiple XAI properties to reflect the full methodological complexity. We aggregated the findings in Tables 3 and 4. Table 3 presents the frequency of significant results found in comparisons of explanations and non-explanations. It provides a general overview of the effectiveness of explanations generated by different XAI methods compared to not using explanations at all. Table 4 compiles the comparison results of different explanations, highlighting their relative performance and indicating which XAI properties tend to outperform others in specific construct groups. Interpreting these results addresses RQ1.2 (i.e., the identification of well-studied and underrepresented XAI properties) and RQ2.1 (i.e., the observation of the effects of different XAI properties on user-centric constructs).
- To complement the quantitative analysis and to reveal context-specific patterns, we conducted a qualitative analysis of each study. We systematically scanned the results, discussion, implications, and conclusion sections of each study for statements describing how the effects of XAI properties vary depending on the factors of user group, task, and domain. We extracted and synthesized these statements to identify recurring patterns regarding the effectiveness of XAI methods and their properties across different user groups and application contexts. This analysis directly addresses RQ2.2 (i.e., how the effects of XAI properties vary across different user groups, tasks, and domains).
- We integrated the findings from the overview, the quantitative analysis, and the qualitative analysis to formulate context-sensitive guidance on selecting XAI methods. This guidance considers the distribution of the evaluated user groups, tasks, domains, and XAI properties in order to minimize potential bias. The resulting guidance links XAI properties to contextual use case demands and supports selecting XAI methods that align with specific goals, such as increasing trust, understanding, or usability. This final step addresses RQ2.3 (i.e., the identification of actionable insights to inform the context-sensitive selection of XAI methods).

Together, these analyses allow us to draw evidence-based conclusions about the effectiveness of XAI properties. They also offer initial recommendations for when and how specific XAI methods can best support human-AI collaboration.

5 Results

This section presents the findings of our systematic review of user studies in the XAI field. We aim to assess the current state of empirical evaluations in user-centric XAI (RQ1) and the ways in which specific XAI properties influence user outcomes across different contexts (RQ2). We start with an overview of the included studies, analyzing the distribution of user groups, tasks, and domains. Next, we report the results of the quantitative analysis, highlighting the significant effects of different XAI properties on user-centric outcomes, such as trust, understanding, usability, and human-AI performance. We then complement these quantitative findings with a qualitative analysis to offer contextual insights into how and why certain XAI properties succeed or fail in specific settings. Finally, we synthesize and connect the results of the analyses to derive empirically grounded recommendations. RQ1 is primarily addressed through the study overview and quantitative analysis, while RQ2 is informed by the insights from the quantitative and qualitative findings.

5.1 Overview of analyzed study contexts

To address RQ1.1 (i.e., to investigate the context of studied use cases and to identify under-represented ones), we first provide an overview of the study settings across the 54 empirical user studies included in our literature review (cf. Table 2). We then categorize each study based on its application domain, task type, and involved user group. This overview highlights the current XAI evaluation focus areas and provides a foundation for identifying underexplored areas and potential biases. The distributions of studies across domains, tasks, and user groups are depicted in Fig. 5. We begin by outlining the key findings addressing RQ1.1.

Key Findings on the Distribution of Study Contexts (RQ1.1)

The current state of user-centric XAI research is characterized by a strong focus on domain-agnostic studies, decision support and classification tasks, and studies involving lay users. Healthcare remains the most frequently represented application domain, while others (e.g., human resources, autonomous driving) have scarcely been studied or are absent altogether. Overall, these findings reveal a focus on general-purpose tasks, lay users, and general domains. This leaves expert users and specialized application areas comparatively understudied. This distribution indicates the need for future user studies that diversify both user groups and task environments to ensure that XAI methods are evaluated across the full range of real-world use cases.

In the following, we provide a more detailed analysis of the study settings.

5.1.1 Domains

The majority of studies are situated in domain-agnostic settings (24 studies), in which the XAI method is evaluated outside a specific application context. Although this approach allows for generalizability, it might overlook important domain-specific factors that influence explanation effectiveness. Among domain-specific studies, healthcare stands out as

Table 2 Included studies, corresponding domains, tasks, and user groups

Study	Domain	Task	User group
Adhikari et al. (2019)	Domain-Agnostic	Classification	Lay Users
Alqaraawi et al. (2020)	Domain-Agnostic	Image Processing	Lay Users
Baniecki et al. (2024)	Healthcare	Classification	Domain Experts
Baron et al. (2025)	Healthcare	Decision Support	Lay Users
Bove et al. (2023)	Finance	Decision Support	Lay Users
Bove et al. (2022)	Customer Support	Decision Support	Lay Users
Buçinca et al. (2020)	Domain-Agnostic	Decision Support	Lay Users
Cheng et al. (2019)	Education	Decision Support	Lay Users
Collaris and Van Wijk (2022)	Domain-Agnostic	Classification	AI Experts
Confalonieri et al. (2021)	Healthcare, Finance	Decision Support	Lay Users
Eisbach et al. (2023)	Human Resources	Decision Support	Lay Users
Famiglini et al. (2024)	Healthcare	Decision Support	Domain Experts
Flutura et al. (2020)	Domain-Agnostic	Image Processing	Lay Users
Gates et al. (2023)	Domain-Agnostic	Decision Support	Lay Users
Guo et al. (2022)	Domain-Agnostic	Evaluation	Lay Users
Hamm et al. (2023)	Domain-Agnostic	Image Processing	Lay Users
Herm et al. (2023)	Healthcare	Decision Support	Lay Users
Hernandez-Bocanegra and Ziegler (2023)	Tourism	Recommender Systems	Lay Users
Jahn et al. (2024)	Domain-Agnostic	Classification	Lay Users
Jalali et al. (2023)	Domain-Agnostic	Classification	AI Experts
Jansen et al. (2024)	Domain-Agnostic	Classification	AI Experts, Lay Users
Karran et al. (2022)	Domain-Agnostic	Image Processing	Lay Users
Koh et al. (2024)	Finance	Decision Support	Lay Users
Kuhl et al. (2023)	Domain-Agnostic	Goal-Driven Simulation	Lay Users
Labarta et al. (2024)	Domain-Agnostic	Image Processing	Lay Users
Lai and Tan (2019)	Tourism	Natural Language Processing	Lay Users
Lee et al. (2023)	Search Engines	Natural Language Processing	Domain Experts
Leichtmann et al. (2023)	Domain-Agnostic	Image Processing	Lay Users
Mertes et al. (2022)	Healthcare	Decision Support	Domain Experts
Naiseh et al. (2023)	Healthcare	Decision Support	Domain Experts
Nourani et al. (2022)	Domain-Agnostic	Image Processing	Lay Users
Panigutti et al. (2023)	Healthcare	Decision Support	Domain Experts
Panigutti et al. (2022)	Healthcare	Decision Support	Domain Experts
Polley et al. (2021)	Search Engines	Natural Language Processing	Lay Users
Qu et al. (2024)	Customer Support	Natural Language Processing	Lay Users
Radensky et al. (2022)	Search Engines	Natural Language Processing	Domain Experts
Righetti et al. (2022)	Domain-Agnostic	Classification	Lay Users
Schrills and Franke (2020)	Domain-Agnostic	Image Processing	Lay Users
Schulze-Weddige and Zylowski (2021)	Sales	Classification	Lay Users
Sovrano and Vitali (2021)	Finance	Decision Support	Lay Users
Stoll et al. (2022)	Customer Support	Natural Language Processing	Lay Users
Suh et al. (2024)	Healthcare	Decision Support	Domain Experts
Szczuka et al. (2024)	Domain-Agnostic	Recommender Systems	Lay Users
Tambwekar and Gombolay (2024)	Domain-Agnostic	Goal-Driven Simulation	Lay Users

Table 2 (continued)

Study	Domain	Task	User group
Tsai et al. (2021)	Healthcare	Decision Support	Lay Users
Van Der Waa et al. (2021)	Healthcare	Decision Support	Lay Users
Vannostrand et al. (2024)	Finance	Decision Support	Lay Users
Wang and Ding (2024)	Customer Support	Classification	Domain Experts
Warren et al. (2024)	Domain-Agnostic	Decision Support	Lay Users
Weitz et al. (2021)	Domain-Agnostic	Speech Recognition	Domain Experts
Weitz et al. (2019)	Domain-Agnostic	Speech Recognition	Lay Users
Wilkinson et al. (2021)	Domain-Agnostic	Recommender Systems	Lay Users
Zhang et al. (2020)	Finance	Classification	Lay Users
Zhong et al. (2019)	Law	Decision Support	Domain Experts

the most frequently studied area (12), followed by finance (6). Other domains such as law, human resources, and tourism are represented sparsely. This distribution indicates a concentration of empirical works in well-established, critical domains such as healthcare, where the demand for transparency and accountability is particularly high. However, it also reveals notable gaps in the evaluation of XAI methods within emerging and possible high-stakes domains like human resources and sales, where following the decision of an AI system can come at high costs. Other domains, such as robotics or autonomous driving, are not represented at all, likely due to their need for specialized explanatory strategies that accommodate real-time and physical-world interactions.

5.1.2 Tasks

Concerning task types, *decision support* constitutes the most common type (22 studies), followed by *classification* (10) and *image processing* (8). The prevalence of decision support and classification tasks reflects the traditional strengths of XAI methods in structured environments (i.e., tasks with structured data). However, tasks involving more interactive, dynamic, or user-driven engagement, such as debugging, complex simulations, and recommendations, are understudied.

5.1.3 User groups

A particularly noteworthy aspect of the identified studies is the distribution of user groups. Most studies (41) evaluated XAI methods with lay users (i.e., individuals without expertise in the respective domain or AI) compared to 11 studies involving domain experts and three studies involving AI experts. This imbalance indicates a bias toward lay users. While focusing on lay users reflects real-world needs in consumer-facing AI applications, the underrepresentation of studies involving expert user groups limits our understanding of how different XAI properties are perceived in professional settings. Consequently, many of the findings from the quantitative analysis may be biased toward general usability and intuitive understanding rather than the requirements of professional or high-stakes environments.

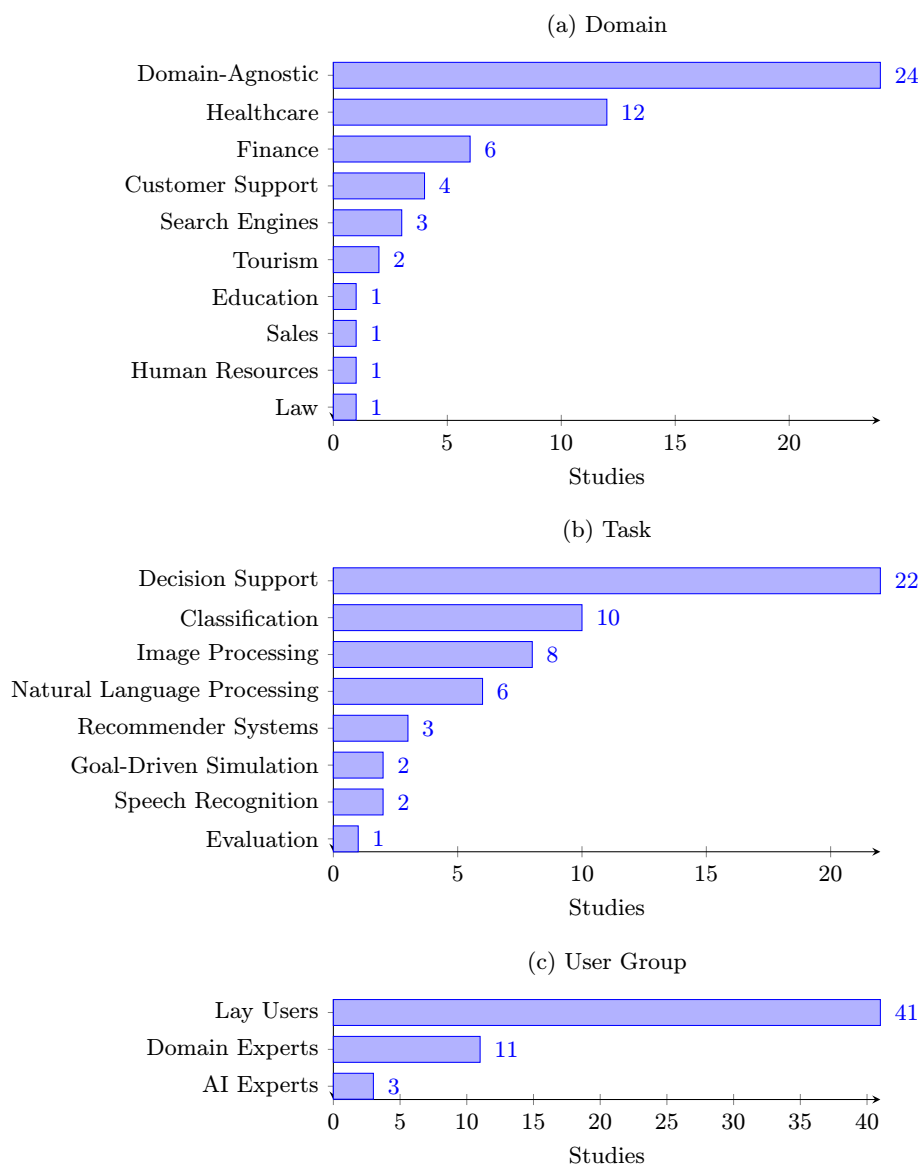


Fig. 5 Distribution of included studies across domains (a), tasks (b), and user groups (c)

5.2 Quantitative analysis

Our quantitative analysis focuses on two key aspects providing insights into the effectiveness of XAI methods. Regarding RQ1.2, which aims to investigate the distribution of evaluated XAI properties and to identify underrepresented ones, we analyze the distribution of evaluated XAI properties when comparing explanations with non-explanations (cf. Table 3). The key insights are summarized as follows:

Key Findings on the Distribution of evaluated XAI Properties (RQ1.2)

Regarding RQ1.2, the findings reveal that empirical research in user-centric XAI has primarily focused on a limited set of XAI properties.

Most studies evaluate local explanations, reflecting a strong emphasis on instance-level interpretability. Concerning result types, feature importance and example-based methods dominate the user studies. For functioning, perturbation-based, leveraging structure, and example search methods are most commonly explored. Regarding output type, most studies focus on visual, textual, or tabular presentations. These findings highlight the concentration of empirical studies on a few XAI properties. The underrepresentation of certain XAI properties (e.g., global, concept-based) highlights important areas for future research to develop a more comprehensive understanding of user-centric XAI.

To investigate the effects of different XAI properties on user-centric constructs (RQ2.1), we analyzed all significant effects observed in the user studies. This analysis accounts for comparisons of explanations to non-explanations (cf. Table 3) and comparisons of different explanations (cf. Table 4), allowing us to contrast the respective patterns. The following summarizes our main findings.

Key Findings on the Effects across User-centric Constructs (RQ2.1)

Compared to non-explanations, our analysis reveals that explanations were predominantly associated with improvements in user-centric outcomes across trust, understanding, usability, and task performance. Trust-related effects demonstrate a clear distinction between implicit and self-reported trust. Implicit trust was often linked to positive effects compared to non-explanations in the studies, whereas for self-reported trust, negative significant effects were reported under certain conditions. This distinction underscores the importance of measuring calibrated rather than increased trust.

Our analysis further reveals recurrent trends across construct categories. Model evaluation methods that use numerical outputs appear to be particularly effective for implicit trust and human-AI performance in the studies. Counterfactual and example-based XAI methods show comparatively strong results with respect to understanding and usability, especially when generated through perturbation or example search and presented in tabular formats. Textual and audio formats were associated with positive effects on self-reported trust and ease of use, whereas visual and feature importance methods yield more mixed results.

Notably, while comparisons to non-explanations underline the value of XAI methods in many contexts, comparisons between different explanations reveal a more nuanced picture. Some methods (e.g., feature importance and visual explanations) perform well against non-explanations but yield mixed or negative results compared to other explanations.

In summary, *explanation effectiveness* is not inherent to individual XAI properties but rather depends on the alignment between XAI properties, user needs and the broader task and domain context. These findings provide empirical support for selecting XAI methods that are tailored to specific user outcomes.

In the following, we present the detailed results of our quantitative analysis with respect to RQ1.2 and RQ2.1.

5.2.1 Comparing explanations to non-explanations

Table 3 summarizes the number of significant effects we observed when comparing explanations to non-explanations. The results are structured around four key dimensions of XAI

Table 3 Results of the studies comparing explanations to non-explanations

	Number of Studies	im-plicit trust	self-reported trust	implicit understanding	self-reported understanding	satisfaction	ease of use	helpfulness	workload	human-AI performance
Scope										
global	3	+1 (1)	- (-)	+1 (1)	+1 (1)	- (-)	- (-)	- (-)	- (-)	+1 (3)
local	44	+10 (10)	+17 (23)	+6 (8)	+26 (28)	+10 (12)	+5 (7)	+13 (13)	+2 (2)	+9 (11)
Result										
example	18	+4 (4)	+10 (14)	+4 (4)	+12 (14)	+2 (4)	+2 (4)	+7 (7)	+1 (1)	+1 (5)
counterfactual	9	+1 (1)	+2 (2)	+3 (3)	+8 (8)	+1 (1)	- (-)	+4 (4)	- (-)	+2 (2)
feature importance	18	+3 (3)	+5 (9)	0 (2)	+13 (15)	+6 (8)	+1 (3)	+5 (5)	+1 (1)	+5 (5)
concepts	2	- (-)	+2 (2)	- (-)	+1 (1)	- (-)	- (-)	+1 (1)	- (-)	- (-)
model statistics	3	+2 (2)	+1 (1)	- (-)	- (-)	- (-)	- (-)	- (-)	- (-)	+2 (2)
data statistics	3	- (-)	+4 (4)	+3 (3)	+2 (2)	- (-)	+3 (3)	+2 (2)	- (-)	- (-)
surrogate model	-	- (-)	- (-)	- (-)	- (-)	- (-)	- (-)	- (-)	- (-)	- (-)
model evaluation	6	+2 (2)	+5 (5)	+3 (3)	+2 (2)	- (-)	+3 (3)	+2 (2)	- (-)	+2 (2)
perturbation	12	+6 (6)	+3 (3)	+3 (3)	+4 (4)	+2 (2)	+1 (1)	+2 (2)	- (-)	+4 (4)
example search	13	+3 (3)	+10 (12)	+3 (3)	+9 (11)	+1 (3)	+2 (4)	+6 (6)	+1 (1)	+1 (5)
model modification	1	- (-)	+1 (1)	- (-)	- (-)	- (-)	- (-)	- (-)	- (-)	- (-)
leveraging structure	12	+1 (1)	+1 (7)	0 (2)	+11 (13)	+3 (5)	+1 (3)	+4 (4)	+1 (1)	+1 (1)
Output type										
no info	9	+1 (1)	+2 (2)	- (-)	+5 (5)	+4 (4)	- (-)	+3 (3)	- (-)	+2 (2)
visual	23	+4 (4)	+8 (12)	+3 (5)	+11 (13)	+4 (6)	+2 (4)	+6 (6)	- (-)	+7 (7)
textual	14	+1 (1)	+6 (8)	+1 (1)	+14 (14)	+6 (6)	+2 (2)	+6 (6)	+2 (2)	+3 (3)
rules	2	+2 (2)	- (-)	- (-)	+1 (1)	- (-)	- (-)	- (-)	- (-)	- (-)
tabular	11	+2 (2)	+9 (9)	+5 (5)	+6 (6)	- (-)	+3 (3)	+5 (5)	- (-)	-1 (3)
numerical	2	+2 (2)	+1 (1)	- (-)	- (-)	- (-)	- (-)	- (-)	- (-)	+1 (1)
model	-	- (-)	- (-)	- (-)	- (-)	- (-)	- (-)	- (-)	- (-)	- (-)
audio	-	- (-)	- (-)	- (-)	- (-)	- (-)	- (-)	- (-)	- (-)	- (-)

Entries indicate the difference between the number of significant positive and negative effects (and the overall number of comparisons) found in the studies compared to the control condition (no explanation)

Table 4 Results of the studies comparing different explanations

	Number of Studies	im- plicit trust	self- reported trust	implicit understanding	self-reported understanding	satisfaction	ease of use	helpfulness	workload	human- AI per- formance
Scope	4	- (-)	- (-)	- (-)	-2 (2)	- (-)	- (-)	-1 (1)	- (-)	0 (4)
	4	- (-)	- (-)	- (-)	+2 (2)	- (-)	- (-)	+1 (1)	- (-)	0 (2)
Result	15	- (-)	+2 (14)	+1 (3)	+5 (11)	+5 (5)	- (-)	+1 (3)	-1 (1)	- (-)
	17	- (-)	-1 (9)	+7 (7)	+8 (12)	+2 (8)	- (-)	+5 (5)	+1 (3)	- (-)
	25	-1 (1)	-3 (7)	-5 (7)	-9 (13)	+1 (13)	- (-)	-3 (5)	+1 (5)	-5 (5)
	2	- (-)	0 (2)	- (-)	- (-)	-1 (1)	- (-)	- (-)	+1 (1)	- (-)
	7	+1 (1)	- (-)	- (-)	- (-)	- (-)	- (-)	- (-)	- (-)	+6 (6)
	4	- (-)	- (-)	- (-)	- (-)	-4 (4)	- (-)	- (-)	- (-)	- (-)
	3	- (-)	- (-)	- (-)	- (-)	-2 (2)	- (-)	- (-)	-2 (2)	0 (2)
Functioning	6	+1 (1)	- (-)	- (-)	- (-)	- (-)	- (-)	- (-)	- (-)	+5 (5)
	17	-1 (1)	+1 (3)	+3 (9)	+1 (11)	+3 (7)	-1 (1)	+2 (4)	+3 (5)	-3 (5)
	8	- (-)	+2 (2)	+1 (3)	+6 (6)	-2 (2)	- (-)	+1 (3)	-2 (2)	+1 (1)
	-	- (-)	- (-)	- (-)	- (-)	- (-)	- (-)	- (-)	- (-)	- (-)
	16	- (-)	-2 (4)	-4 (8)	-8 (12)	-4 (6)	0 (2)	-2 (4)	-4 (4)	-4 (4)
	6	- (-)	0 (6)	- (-)	-2 (2)	0 (4)	- (-)	-1 (1)	0 (2)	0 (2)
Output type	26	-1 (1)	-1 (17)	-1 (3)	-4 (10)	+4 (6)	-1 (1)	0 (4)	+2 (6)	-4 (6)
	13	- (-)	+5 (7)	- (-)	+1 (3)	-1 (3)	+2 (2)	- (-)	+1 (1)	0 (2)
	3	- (-)	- (-)	- (-)	- (-)	- (-)	0 (2)	- (-)	- (-)	-2 (2)
	8	- (-)	+1 (3)	+2 (2)	+5 (7)	+1 (1)	- (-)	+1 (3)	-1 (1)	- (-)
	16	+1 (1)	-6 (6)	- (-)	- (-)	+3 (7)	- (-)	- (-)	+4 (4)	+5 (5)
	7	- (-)	- (-)	- (-)	- (-)	-2 (2)	-1 (3)	- (-)	-2 (2)	+1 (3)
	2	- (-)	+2 (2)	- (-)	- (-)	- (-)	- (-)	- (-)	- (-)	- (-)

Entries indicate the difference between the number of significant positive and negative effects (and the overall number of comparisons) found in the included studies compared to the control condition (different XAI property)

properties: scope, result, functioning, and output type (cf. Speith (2022)). For each property, we report the tendency of significant effects (i.e., the number of positive significant effects minus the number of negative significant effects) observed across the key construct clusters of trust, understanding, usability, and human-AI performance as well as the overall number of comparisons found in the reviewed studies.

As a fundamental finding of this analysis, explanations were predominantly associated with positive effects compared to the absence of explanations. Across most constructs, introducing explanations improved user outcomes. Local explanations, example-based, and feature importance methods, perturbation methods and example search methods, as well as visual, textual, and tabular output types were mostly linked to positive effects across trust, understanding, usability, and human-AI performance. These results indicate the value of integrating explanations into AI systems and provide empirical support for the use of XAI methods in practice.

A noteworthy observation relates to the trust construct cluster. No statistically significant negative effects were reported for *implicit trust* measures, such as appropriate reliance or trust calibration. In contrast, few studies reported negative effects for *self-reported trust*. This distinction is meaningful as self-reported trust does not always align with desirable outcomes. In fact, overtrust might occur when users place unwarranted confidence in AI systems, whereas implicit trust captures behaviorally grounded responses. For example, they determine whether users appropriately follow decisions provided by an AI system based on its correctness. Therefore, implicit trust reduces the risk of misinterpreting high trust as inherently positive. Furthermore, no statistically significant negative effects were recorded for the construct categories *helpfulness* and *workload*. For the other categories, negative effects, when reported, are sparse and are not concentrated on specific XAI properties. These findings indicate that explanations were not associated with statistically significant increases in workload or reductions in helpfulness and implicit trust. However, this absence of reported negative effects must not be interpreted as evidence that such adverse effects cannot occur in other contexts.

The comparison of explanations and non-explanations reports on the frequency with which XAI properties are evaluated in user studies. Consequently, Table 3 depicts the distribution of XAI properties commonly evaluated in literature. This highlights the areas on which empirical research attention has focused to date. Concerning *scope*, the majority of studies applied *local* explanations (44 studies), which focus on individual predictions. Only a few studies investigated *global* explanations (3), reflecting the broader focus on local interpretability in XAI research.

For the *result* category, *feature importance* and *example-based* methods were the most frequently studied (18 each), followed by *counterfactuals* (9). Other XAI properties, such as *concepts* and *model statistics*, were sparsely studied (cf. Table 3). This indicates a gap in current research coverage.

In the *functioning* category, most XAI methods operated via *perturbation* (12), *example search* (13), or *leveraging structure* (12). Only a few studies used *model evaluation* (6) and *model modification* (1). This is likely because such XAI methods often yield model-specific XAI properties, which were excluded from our review. Notably, nine studies did not specify any functioning principle, underscoring the need for improved methodological transparency in XAI reporting.

Concerning *output type*, *visual* explanations (23) were the most frequently evaluated and were predominantly associated with positive effects across all construct clusters. *Textual* (14) and *tabular* (11) outputs were also frequently evaluated. However, other output types (e.g., *rules* (1), *numerical outputs* (2)) were underrepresented. These findings reveal a preference for visual, textual, and tabular outputs, as well as a research gap in underrepresented XAI properties.

5.2.2 Comparing different explanations

The following findings concern studies that compare different explanations based on the underlying XAI properties and examine how these influence the construct clusters. Table 4 summarizes the tendency of significant effects (i.e., the number of positive significant effects minus the number of negative significant effects) observed across the construct clusters of *trust*, *understanding*, *usability*, and *human-AI performance* when comparing explanations and the overall number of comparisons found in the reviewed studies. We report tendencies based on positive or mixed outcomes, emphasizing which XAI properties show comparatively favorable patterns in particular areas.

Only one study reported significant results for *implicit trust*, making it difficult to draw generalizable conclusions. In contrast, *self-reported trust* was assessed more frequently, yielding mainly mixed results. Among the evaluated properties, textual output stands out with a clearly positive tendency (tendency: +5 (number of comparisons: 7)), whereas numerical output was consistently associated with negative effects (-6(6)).

Regarding *implicit* and *self-reported understanding*, XAI methods that generate counterfactuals (implicit: +7(7); self-reported: +8(12)) or example-based methods (+1(3); +5(11)) showed comparatively strong performance across the studies. These explanations appeared particularly effective when being produced through example search (+1(3); +6(6)) and presented in tabular format (+2(2); +5(7)). In contrast, feature importance methods (-5(7); -9(13)) and leveraging structure methods (-4(8); -8(12)) were more frequently associated with negative effects. The remaining properties yielded mixed results.

Regarding usability, which encompasses *satisfaction*, *ease of use*, *helpfulness*, and *workload*, the results are generally mixed and appear highly context-dependent. Example-based methods and visual outputs (+4(6)) showed particularly favorable results concerning satisfaction (+5(5)). Regarding helpfulness, counterfactuals stood out with consistently positive effects (+5(5)). Notably, numerical outputs were associated with reduced workload (+4(4)).

The most positive effects on human-AI performance were observed for explanations produced by model evaluation (+5(5)), resulting in model statistics (+6(6)) and presented in numerical form (+5(5)). In contrast, feature importance methods (-5(5)), generated by leveraging structure (-4(4)) and presented in visual form (-4(6)), were frequently associated with negative effects. Other XAI properties showed no clear patterns and yielded mixed results.

Comparing different explanations across user-centric constructs indicates that explanation effectiveness is highly context-dependent. While certain XAI properties performed predominantly well across constructs, others showed mixed results. This indicates that their success may vary significantly depending on the respective user group, task, or domain in which they are applied.

In the studies, *counterfactuals*, *perturbation-based* functioning, and *tabular* or *numerical* outputs emerged as promising candidates for improving understanding, trust, and usability in many scenarios.

At the same time, most XAI properties (e.g., example-based, feature importance, textual or visual presentations) show mixed or balanced effects. This variability does not indicate poor performance in general; rather, it indicates that these methods may be sensitive to contextual factors, such as the user's level of expertise, the cognitive demands of the task, and the nature of the decision domain. Textual explanations, for instance, may foster ease of use for lay users in low-stakes contexts, but could be perceived as being vague or insufficient by experts in high-stakes domains.

5.2.3 Cross-comparison of study results

Comparing explanations with non-explanations as well as with alternative explanations provides a dual perspective that reveals important insights and potential pitfalls in interpreting the effectiveness of XAI methods.

Comparisons between explanations and non-explanations confirm that providing explanations is predominantly associated with improved user-centered outcomes across *trust*, *understanding*, *usability*, and *human-AI performance*. This further supports the benefit of integrating explainability into AI systems. However, the results also reflect the uneven distribution of research attention across XAI methods. More frequently studied methods are more often represented in positive effect counts in these comparisons. To overcome this challenge, analyzing comparisons between different explanations provides a more nuanced assessment of XAI properties. This analysis can better identify areas where certain XAI properties show stronger results under comparable conditions.

A particularly illustrative example is *feature importance* methods. Compared to non-explanations, they are associated with improvements in all four construct clusters (trust: tendency: +8 (number of comparisons: 12); understanding: +13(17); usability: +13(17); human-AI performance: +5(5)). However, when compared directly to other explanations, they produce negative and mixed results for the construct clusters (-4(8), -14(20), -1(23), -5(5)). The case of visual explanations further demonstrates the need for nuanced interpretation. Visual formats were predominantly associated with positive outcomes compared to non-explanations (e.g., self-reported trust: +8(12), human-AI performance: +7(7)). However, they exhibit a balanced distribution of positive and negative outcomes when compared to other explanations (-1(17), -4(6)). This variability implies that the effectiveness of visual explanations depends on their design and the context in which they are applied.

Overall, our results demonstrate that broad claims about the effectiveness of explanations must be made with caution. While comparisons of explanations and non-explanations indicate the general benefit of transparency in AI systems, comparative evaluations are necessary to inform XAI method selection.

5.3 Qualitative analysis

Due to the imbalanced distribution of user groups in our dataset (i.e., studies), particularly the prevalence of studies involving lay users, we supplement our quantitative

findings with a qualitative analysis to address RQ2.2. This analysis explores how XAI property effects vary across different use case contexts.

The analysis is based on two dimensions: *user characteristics* (i.e., AI expertise and domain expertise) and *task- & domain-specific contexts* (e.g., complexity, stakes, application type). This section synthesizes insights from the studies to describe patterns that inform the context-sensitive design and application of XAI methods. In the following, we highlight the main findings of our qualitative analysis.

Key Findings on Context-dependent Effects of XAI Properties (RQ2.2)

The qualitative analysis reveals that the effectiveness of XAI methods strongly depends on user characteristics and the context of the use case. Across the studies, lay users were more likely to benefit from low-complexity explanations, such as local, example-based, or counterfactual formats, presented by familiar outputs that minimize cognitive load. In contrast, domain experts and AI experts tended to prefer explanations concerning feature importance or model statistics, especially when output types were used that fit to domain-specific knowledge or established workflows of the users.

Additionally, contextual factors such as task complexity and application stakes appear to substantially influence the effectiveness of explanations. In high-stakes domains like healthcare, fostering appropriate trust is crucial. Therefore, informative methods such as model evaluations were associated with positive outcomes. Furthermore, the reviewed studies suggest that local explanations are well suited for a wide variety of tasks, while global explanations appear particularly useful for tasks like model debugging or system-level inspection.

Interactivity and customization were associated with positive outcomes by allowing users to tailor explanations to their needs, thereby fostering both trust and understanding. Additionally, integrating both contextual and domain knowledge (e.g., ontologies and narratives) may improve interpretability and decision quality.

These findings underscore the importance of a context-sensitive approach to select XAI methods. They indicate that XAI methods should be carefully matched to users' expertise, task demands, and domain-specific requirements to support effective outcomes.

In the following, we provide a detailed description of the results of our qualitative analysis with respect to RQ2.2.

5.3.1 Aligning explanation information with user characteristics

Several studies emphasize the importance of tailoring the provided information to the users' knowledge (cf. Miller (2019); Herm et al. (2023); Hernandez-Bocanegra and Ziegler (2023); Jansen et al. (2024); Mertes et al. (2022)). Karran et al. (2022) describe this alignment as the congruence between the problem-solving task and its representation, ensuring a fit with the mental model of the user and task-specific knowledge.

Across the studies, lay users tended to benefit from explanations with lower cognitive demands due to their lack of familiarity with the domain, task, and AI systems involved. For example, Herm et al. (2023) evaluated five XAI methods in healthcare and concluded that local explanations were more effective for lay users as they impose a lower cognitive load than global explanations. This finding is supported by Hamm et al. (2023), who observed similar advantages of local explanations for lay users in the context of an image-processing task.

Moreover, several studies have demonstrated the effectiveness of example-based methods and counterfactuals for lay users (cf. Adhikari et al. (2019); Warren et al. (2024)). These findings highlight the importance of minimizing the cognitive load to help lay users to comprehend explanations, as counterfactual and example-based methods tend to be more intuitive and accessible to human understanding (Miller 2019). Another study compared the

effects of explanations presented by a graphical user interface (GUI) with those presented by a natural language processing (NLP) agent that could be accessed via dialog (Hernandez-Bocanegra and Ziegler 2023). Again, the authors observed that the less demanding alternative performed better. They claim that the NLP agent requires more cognitive load as it necessitates formulating questions to obtain the desired explanations.

Wang and Ding (2024) emphasize the importance of selecting the right output type for lay users. This finding is supported by Jansen et al. (2024), who compared two explanations with the same amount of information. One explanation was visualized with a card design to ease its comprehension. Again, positive effects were observed when the explanation required less cognitive load. In this regard, Vannostrand et al. (2024) studied the comparison presenting single-instance counterfactuals and region counterfactuals. In the case of region counterfactuals, the explanation does not provide alternative feature values. Instead, it provides alternative feature ranges classified differently from the original input. The authors observed that the region counterfactuals performed better for lay users. This aligns with findings of Warren et al. (2024), who propose transforming continuous variables into categorical ones (i.e., by mapping value ranges to categories) in order to enhance understanding. Schulze-Weddige and Zylowski (2021) propose presenting explanations in an output type familiar to the user. When no specific information about the user group is available, they recommend using well-known forms of presentation (e.g., bar plots). Furthermore, to ensure that the output type fits well to the respective user group, Herm et al. (2023) propose a UCD process. For lay users, the presentation plays a pivotal role. However, for decision-makers with high task-related capability (i.e., AI experts and domain experts), the information provided in explanations is more important (Wang and Ding 2024).

In line with the paradigm of tailoring the information of an explanation to the user's knowledge level, both AI experts and domain experts benefit from more detailed explanations. For example, Mertes et al. (2022) observed that counterfactuals emphasizing substantial changes in critical features, designed to ensure a shift in the predicted class, can be effective for lay users. However, such explanations may be less suitable for experts as they lack detailed information required to provide novel insights (Jahn et al. 2024). To enable the latter, Jahn et al. (2024) propose using XAI methods that highlight unexpected details. For learned workforce (i.e., domain experts and AI experts), Herm et al. (2023) and Warren et al. (2024) suggest that using causal and coefficient-based methods (e.g., feature importances) may lead to better performance while remaining sufficiently explainable. This aligns well with Collaris and Van Wijk (2022) and Flutura et al. (2020), who emphasize the effectiveness of feature importance methods for AI experts.

Regarding the output type, we need to differentiate more clearly between AI experts and domain experts. In their study, Guo et al. (2022) observed that output types perceived as intuitive by AI experts were perceived as the opposite by domain experts. As proposed by Schulze-Weddige and Zylowski (2021), familiarity with the output type plays a key role. The familiarity of AI experts with diverse information presentation styles may reduce their preference for textual explanations (Tambwekar and Gombolay 2024). In contrast, domain experts are typically less familiar with different output types of explanations. In their study, Famiglini et al. (2024) compared different coloring styles for class activation maps. Although a newly proposed coloring style provided more information, healthcare professionals preferred the traditional style, which may be attributed to their existing familiarity with this presentation. Therefore, using output types the intended user group is famil-

iar with may facilitate more effective interpretation. When this cannot be inferred, Guo et al. (2022) propose a UCD process to find a suitable output type.

5.3.2 Complexity, stakes, and tailored applications

User characteristics as well as task- & domain-specific factors influence the effectiveness of different XAI methods (Hamm et al. 2023). For example, Jahn et al. (2024) argue that highlighting feature values deviating from expected norms can be beneficial for familiar tasks and domains (e.g., professionals working with decision support systems in healthcare or manufacturing). Considering these factors, it is important to evaluate XAI methods in the context of proxy tasks being close to the real task these methods will be used for (Bućinca et al. 2020). While Gates et al. (2023) observed that all other types of explanations were preferred over the combination of example-based methods and counterfactuals in a blood alcohol prediction task, they suggested further evaluating these methods in the context of more complex tasks and domains.

However, complexity is not the only relevant factor; the stakes of a task and its domain constitute important considerations as well (Baron et al. 2025). Low-stakes applications might benefit from explanations that reduce cognitive effort and encourage its usage by potentially unmotivated users, as observed by Vannostrand et al. (2024). In contrast, high-stakes applications (e.g., those in the healthcare domain) require establishing appropriate trust (Panigutti et al. 2023). The importance of appropriate trust in high-stakes contexts is emphasized by the work of Leichtmann et al. (2023) as well. The authors argue that the performance of an AI system directly affects the perception of XAI methods; low-performing systems tend to reduce perceived trust, a critical factor for trust calibration, especially in high-stakes contexts (Leichtmann et al. 2023).

In addition to complexity and stakes, the qualitative analysis revealed findings that may improve the effectiveness of explanations in more specific use cases. For example, presenting the AI system's confidence scores was found to help users to identify its decision boundaries (Kuhl et al. 2023). Especially, this is beneficial for complementary performance when the AI system and its users have different decision boundaries (Kuhl et al. 2023). Furthermore, for tasks such as model debugging, in which the user needs to understand how the AI system functions rather than to scrutinize individual decisions, local explanations may be insufficient for providing this information (Alqaraawi et al. 2020). In such use cases, Alqaraawi et al. (2020) and Hamm et al. (2023) propose the use of *global explanations*. In contrast, NLP tasks such as explaining search engine results might benefit from combining global and local explanations (Radensky et al. 2022). Other use cases might benefit from a concise presentation of knowledge (Righetti et al. 2022). In such cases, explaining via formula rather than showing model representations can be beneficial (Righetti et al. 2022). In applications with clear goals (e.g., improving the outcome of a production factory), Kuhl et al. (2023) observed that counterfactuals can help identify alternative inputs leading to the desired outcomes.

Beyond context-dependent effects, the qualitative analysis reveals more broadly applicable insights. The most frequently recommended approaches for improving XAI methods are *interactivity* and *customization* (Cheng et al. 2019; Guo et al. 2022; Hamm et al. 2023; Hernandez-Bocanegra and Ziegler 2023; Koh et al. 2024; Tsai et al. 2021; Weitz et al. 2021). These XAI properties enable users to adjust the XAI method to their needs, prevent-

ing cognitive overload and providing the opportunity to request more detailed information (Koh et al. 2024; Naiseh et al. 2023). Hamm et al. (2023) observed that manifold interaction options with the XAI method are perceived as beneficial.

Regarding output type, Schulze-Weddige and Zylowski (2021) indicate the importance of avoiding misinterpretations. The proposals for ensuring correct interpretations of explanations have been varied. For example, Qu et al. (2024) observed that feature importance methods often require additional explanation as their predictive power might not be intuitively apparent to users. In line with this, Weitz et al. (2021) propose using additional information in the form of other output types to help users to become familiar with the provided visual explanation. Karran et al. (2022) recommend utilizing context, i.e., the input of the AI system, in addition to the explanation. In order to enhance understanding and trust, Confalonieri et al. (2021) and Labarta et al. (2024) found that providing relevant domain and common knowledge (e.g., in the form of ontologies and narratives) was beneficial. Finally, Nourani et al. (2022) propose a setup to familiarize users with an emerging XAI method. First, this setup shows cases with correct AI decisions, followed by cases with incorrect ones. In such a setting, explanations might reveal strengths as well as weaknesses of the system, which may support appropriate trust (Nourani et al. 2022).

5.4 Empirically grounded guidance for selecting XAI methods

By connecting the quantitative and qualitative findings, we derive guidance for selecting use case-specific XAI properties. This answers RQ2.3, which aims to identify actionable insights that guide the context-sensitive selection of XAI methods. The main results are summarized below (see also Fig. 6). Note that the guidance (cf. Fig. 6) does not represent universally optimal design choices. Instead, it summarizes recurring patterns observed across the reviewed studies indicating empirically derived tendencies that may vary depending on the respective application context.

Key Findings on Context-sensitive XAI Method Selection (RQ2.3)

Integrating the results of the quantitative and qualitative analyses allows us to derive actionable guidance for selecting XAI methods based on user and use case characteristics (cf. Fig. 6). Across the studies, several consistent tendencies can be observed for the XAI property dimensions.

Regarding *scope*, only a few comparisons were reported in the studies. Nevertheless, the qualitative analysis indicates that local explanations were frequently associated with positive outcomes and appeared to be suitable for many use cases. Global explanations turned out to be particularly useful for specialized tasks requiring system-level insights, such as model debugging.

To identify a suitable *result* type, the qualitative analysis indicates the importance of distinguishing between lay users and expert users. In the quantitative analysis we observed that counterfactuals stand out for their strong contributions to *understanding* and *usability*, especially for lay users. Nevertheless, from results of the qualitative and quantitative findings we can conclude that more detailed counterfactuals can align with the needs of AI experts and domain experts as well. Furthermore, example-based methods were more frequently associated with increased trust. Combined with the qualitative insights, these explanations appear especially suitable for lay users. The quantitative analysis further indicates that model statistics were particularly effective for improving human-AI performance and implicit trust. This pattern is supported by results of the qualitative analysis, especially with respect to expert users. The qualitative analysis further indicates that feature importance methods may be effective for domain experts and AI experts who both require detailed analytical information.

Though often underreported, the functioning of XAI methods plays a crucial role as well. Example search methods produce realistic instances, which appear to be suitable especially for lay users. In contrast, perturbation-based methods provide analytical flexibility often appreciated by expert users. Methods based on model evaluation contribute to improved human-AI performance.

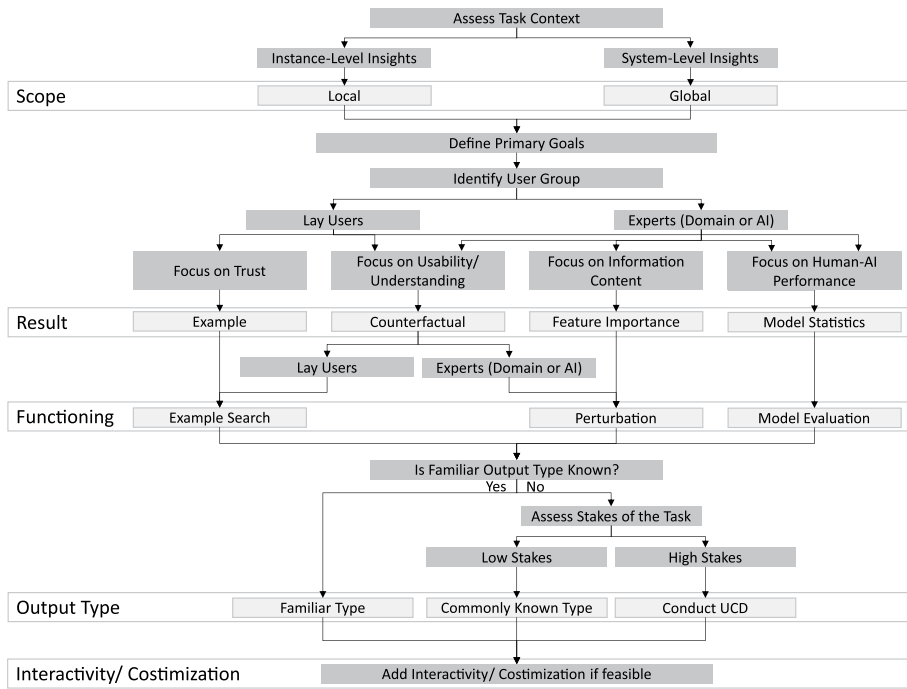


Fig. 6 Selection Guidance for the different XAI properties. Note that this is a brief overview and does not reflect the full methodological complexity. For more details, we refer to Sect. 5

Regarding output type, the quantitative analysis largely shows mixed results. Nevertheless, the qualitative findings indicate that practitioners may benefit from selecting output types that are familiar to the intended user group. If familiarity cannot be inferred, practitioners may either rely on commonly known formats or conduct a UCD process, depending on the stakes of the task. Finally, incorporating *interactivity* and *customization* appears to be broadly advantageous, empowering users to adapt explanations to their cognitive and task-specific needs.

While these results summarize the key insights, in the following we describe our findings in more detail, including additional considerations for selecting appropriate XAI methods for specific use cases.

Regarding *scope*, the majority of use cases benefit from local explanations that focus on a specific outcome, as observed in quantitative and qualitative analyses. The qualitative analysis indicates that global explanations may be particularly effective for applications that require system-level insights.

From the quantitative analysis, we can conclude that counterfactuals showed comparatively strong performance with regard to understanding, helpfulness, and workload, whereas example-based methods tended to be particularly effective in fostering self-reported trust and satisfaction. Together with the qualitative findings, this underscores the importance of these XAI methods for lay users. While the quantitative analysis has delivered poor results for feature importance methods, the qualitative findings indicate that these methods may be particularly effective for AI experts and domain experts. Additionally, the qualitative analysis revealed that model statistics can help users to calibrate trust, and thereby, to increase human-AI performance, as demonstrated in the quantitative analysis. The qualitative analy-

sis further indicates that model statistics were particularly effective in identifying cases of high uncertainty, i.e., situations in which the AI system exhibits low confidence in its decision-making.

A similar pattern applies to XAI methods that are based on model evaluation. Additionally, we found that the functioning of XAI methods received limited attention in the studies. Several studies did not provide information on the functioning of the applied XAI method. This distinction appears meaningful, as different functioning properties may lead to different characteristics of the explanations. For instance, example search produces realistic explanations, as the examples are drawn from real data points (i.e., the training data). However, the limited number of samples in the training data constrains the ability of example search methods to identify instances with the desired characteristics. In contrast, perturbation-based methods may generate unrealistic explanations by varying the model input to identify data points that might not occur in real-world data. The quantitative analysis reveals that perturbation-based, model evaluation, and example search methods were linked to favorable outcomes across multiple constructs. In line with cognitive load theory, example search methods may be particularly appropriate for lay users to ensure realistic data. Additionally, perturbation-based methods might be suitable for expert users, as they are capable of generating explanations with the desired details.

The quantitative analysis reveals mixed results regarding output types. However, textual and tabular explanations show comparatively positive tendencies across several constructs. Both formats are widely used and generally familiar to most users. This observation aligns with the qualitative findings, which emphasize the importance of selecting output types with which users are familiar. While it cannot be assumed that users are familiar with output types such as rules, numerical information, or model representations, textual and tabular formats represent broadly recognized ways of presenting information. Consequently, the quantitative and qualitative findings jointly indicate the effectiveness of output types that are familiar to the intended user group. As suggested by Guo et al. (2022) and Herm et al. (2023), low-stakes applications may consider using a commonly known output type if the user group's familiarity with output types cannot be ensured. For high-stakes applications, a UCD process may be particularly valuable for identifying an appropriate output type.

The qualitative findings further indicate that incorporating interactivity and customization into XAI interfaces is a broadly applicable strategy that fosters usability and supports diverse user preferences. Even when not actively used, interactive features can enhance the perceived utility of the XAI method (Hamm et al. 2023). Moreover, customization enables users to adjust complexity according to their cognitive capacity (Koh et al. 2024).

6 Discussion

6.1 Theoretical implications

The findings of the quantitative and qualitative analyses contribute to the theoretical development of user-centric XAI by clarifying how specific explanation properties relate to user-centered outcomes across different contexts. While prior research has emphasized the importance of aligning explanations with user characteristics and task requirements (Miller 2019; Karran et al. 2022; Herm et al. 2023), our synthesis provides empirical evidence on

how this alignment manifests in practice. Across the studies, explanation effectiveness varies depending on the interaction between explanation properties (e.g., scope, explanation type, or output format), user expertise, and contextual factors such as task demands or application domains. These findings reinforce the growing consensus that there is no universally optimal explanation strategy for AI systems. Instead, effective explanation design emerges from the contextual alignment of XAI properties with the informational needs and cognitive capabilities of users.

Building on this perspective, our review advances the empirical and theoretical understanding of explanation design in *user-centric XAI* in the following aspects.

1. Our analyses highlight structural imbalances in the empirical landscape of user-centric XAI research and their implications for theory development in this field. Although substantial progress has been made in evaluating specific XAI methods and user groups, the landscape of empirical research in XAI still lacks diversity. Specifically, the XAI methods and user groups involved in existing user studies only represent a limited subset of the broader XAI domain. For example, while many studies analyze feature importance, counterfactual, and example-based methods, other explanation approaches, such as concept-based or surrogate model methods, remain comparatively underexplored in empirical user studies. From a theoretical perspective, this concentration of empirical attention may unintentionally shape the understanding of explanation effectiveness around a narrow subset of XAI approaches. These findings therefore indicate that current theoretical insights about user-centric XAI are grounded on a limited empirical base. Expanding empirical investigations into underexplored XAI methods, user groups, and use case contexts therefore becomes necessary to broaden the theoretical understanding of how XAI properties influence user outcomes across diverse settings.
2. Our work provides empirically grounded guidance for selecting XAI methods based on patterns observed across 54 user studies. In particular, we identified XAI properties that tend to align with the needs of specific user groups and application contexts. Furthermore, these insights allow us to further examine and refine existing theoretical frameworks by grounding them on empirical evidence. For instance, our results provide empirical support for the use of counterfactuals for lay users and, in certain specialized cases, for experts. Additionally, as proposed by Hashemi et al. (2025), our results indicate that feature importance methods are more suitable for expert users, whereas the reviewed studies indicate that global explanations appear particularly relevant for model debugging tasks. Moreover, we identified aspects that challenge previous recommendations. Hashemi et al. (2025) propose using numerical representations to increase accuracy and visualizations for the sake of an improved comprehension. However, the qualitative analysis indicates that the effectiveness of output types is strongly influenced by the familiarity of the user with the respective type. In many scenarios, developers can reasonably infer preferred output types based on characteristics of the target user groups. However, given the observed variability in user preferences across contexts, a UCD process may help identify the most suitable output. To a large extent, our findings are consistent with the question-based framework proposed by Liao et al. (2020), which emphasizes aligning explanation design with user intent and task goals. Across all studies, explanation effectiveness depended strongly on user expertise, task type, and contextual demands, providing empirical support for this user-centered design perspective.

However, our results extend the framework provided by Liao et al. (2020) by demonstrating that the effectiveness of answering specific user questions is mediated by underlying XAI properties, such as scope, functioning, and output format. Thus, while Liao et al. (2020) provide a conceptual foundation for question-driven explanation design, our synthesis offers empirical insights on how specific XAI properties operationalize this alignment in practice. Our findings complement the construct-based evaluation perspective introduced by Rong et al. (2023), who propose clustering user-centric outcomes into constructs such as trust, understanding, usability, and performance. The quantitative analysis supports this multidimensional perspective and further demonstrates that these constructs respond differently to distinct XAI properties. For instance, model statistics and evaluation-based explanations appear particularly effective for implicit trust measures and human-AI performance, whereas counterfactuals and example-based methods more consistently support understanding and usability. This differentiation provides empirical granularity to the taxonomy proposed by Rong et al. (2023), underscoring the importance of aligning XAI selection with specific user-centric goals rather than assuming uniform effects across constructs. Taken together, these findings contribute to the theoretical development of user-centric XAI by empirically linking explanation design properties with user-centric outcomes, thereby bridging conceptual frameworks with empirically grounded guidance for XAI method selection.

3. The qualitative analysis yields two key insights that reinforce existing social science theory on designing explanations that align with user needs. Specifically, the analyzed studies provide consistent evidence of a systematic relationship between the users' knowledge level and their preferred explanation characteristics. Across the reviewed studies, lay users tended to prefer intuitive, high-level explanations (e.g., example-based methods or simple counterfactuals), whereas domain experts and AI experts tended to expect more detailed and analytical information about the decision-making process (e.g., advanced counterfactuals or feature importance methods). This pattern indicates that explanation design should balance cognitive accessibility and informational depth depending on the expertise of the target user group. Furthermore, the analyses highlight the importance of user familiarity with output types. The studies indicate that users tend to prefer interacting with familiar formats when trying to comprehend complex information. Consequently, interpretability may not only be influenced by the informational content of explanations but also by their alignment with users' existing mental models.
4. Regarding human-grounded evaluation, our findings underscore the importance of distinguishing between implicit and self-reported measures of trust, and reinforce the theoretical distinction between the two (cf. Hoff and Bashir (2015)). We observe divergent effects of explanations across these two dimensions. Self-reported trust showed mixed results, while implicit trust measured by behavioral indicators was consistently associated with positive significant effects when explanations were provided. This pattern indicates that explanations may be more effective in improving implicit trust in AI systems than in increasing self-reported perceptions of trust. Considering the prevalence of self-reported trust measures in comparisons of different explanations, our findings highlight the importance of evaluating XAI methods based on calibrated trust rather than solely focusing on increasing trust.

6.2 Practical implications

Our results offer actionable insights for practitioners and system designers who want to implement XAI methods in real-world applications. Most notably, our findings illustrate that no single XAI method consistently outperforms the others. Rather, the effectiveness of an XAI method depends heavily on contextual factors, including user expertise, task complexity, domain familiarity, and application stakes.

Considering this dependency, the combination of quantitative and qualitative analyses provides practical guidance in the field of user-centric XAI. Specifically, our work can help practitioners to design more effective XAI methods that meet the needs of end users by providing empirically driven recommendations on selecting appropriate XAI methods based on contextual factors. This guidance can support the development of XAI methods that are not only technically sound but are optimized for specific XAI goals, such as trust, understanding, usability, and human-AI performance with regard to a respective domain and application as well. Our guidance is intended for researchers who are working on novel XAI methods or conducting further empirical studies as well as practitioners selecting appropriate XAI methods for specific applications.

6.3 Limitations and future work

Despite the breadth of the presented review, there are several limitations that need to be discussed.

By requiring statistically significant results as an inclusion criterion, our review does not incorporate purely qualitative studies or studies reporting exclusively non-significant results. While such studies may provide valuable contextual insights regarding the qualitative analysis, this decision reflects the focus on studies that enable a structured aggregation of positive and negative effects. As a consequence, several types of insights remain outside our synthesis. First, by excluding purely qualitative studies, our qualitative analysis does not incorporate the additional contextual insights and conclusions that such studies could have contributed. Second, non-significant results may convey nuanced observations that could support further conclusions, which our synthesis does not consider. Third, even within the studies that we include, more detailed observations, such as subtle differences between the reported effects within a study, remain unanalyzed, as we focused on the structured aggregation of effects rather than on detailed within-study patterns. Future research could complement this perspective by systematically integrating qualitative findings to provide a more comprehensive understanding of contextual influences on the effectiveness of XAI properties.

We did not conduct a formal risk-of-bias or methodological quality assessment of the included studies. As a result, the synthesized findings do not differentiate between studies of varying experimental rigor, sample size, or statistical power. This reflects our decision to focus on identifying recurring patterns of statistically significant effects across the empirical XAI literature rather than weighting studies according to predefined quality criteria. As contextual factors of the included studies are not reflected in the aggregated findings, the robustness of our synthesized conclusions may be affected. Future systematic reviews could extend this work by incorporating structured quality assessment frameworks or weighting

schemes to examine how study design characteristics influence the robustness and generalizability of reported effects.

A further methodological consideration concerns the coding procedure underlying our analysis. As described in Sect. 3.3, data extraction and categorization were primarily conducted by a single researcher, with partial validation through random checks by two additional researchers and consensus-based resolution of ambiguities. The procedure does not include independent double-coding of all included studies and does not yield formal inter-rater agreement metrics. Given the interpretative nature of several classification decisions, such as mapping of measured constructs to construct clusters, this approach represents a potential source of bias. Future systematic reviews could strengthen reliability by adopting independent multi-coder schemes and reporting concordance metrics.

Although the analyses provide insights relevant for various user groups, the distribution of the studies across these user groups is uneven. As mentioned, a large proportion of the evaluated studies focuses on lay users, leaving both domain experts and AI experts underrepresented. The qualitative analysis helps bridge this gap by synthesizing insights from the smaller number of studies involving expert users. Nevertheless, the generalizability of the quantitative findings across diverse user groups remains limited, particularly in professional and high-stakes contexts. Considering this limitation, we encourage future research to prioritize empirical user studies that include domain experts or AI experts to ensure a more balanced and comprehensive understanding of explanation effectiveness across user types.

We classified and analyzed a wide range of XAI methods. However, some of them (e.g., surrogate models, concept-based explanations) remain severely underrepresented. This might be due to limited adoption, recent development, or applicability restricted to specific model types. To address this issue, we grouped XAI methods into four categories according to their XAI properties, which allows for more inclusive generalizations across method families. However, this abstraction comes at the cost of more detailed insights into emerging or niche approaches. Future research should broaden the scope of empirical evaluations to include these underrepresented XAI methods as they become more widely adopted in specialized domains.

Due to the emphasis on generalizable results, certain domains (e.g., robotics, autonomous driving) remain underrepresented as research in these domains tends to rely on model-specific XAI methods. Our analysis addresses this limitation by focusing on XAI properties. This enables us to include XAI properties that are abstracted from model-specific XAI methods. Nevertheless, domain-specific constraints are likely to systematically shape explanation effectiveness. Domains differ in temporal demands, accountability structures, workflow integration, and the physical or abstract nature of decision contexts. For instance, real-time domains such as robotics or autonomous driving impose strict latency constraints that may limit the feasibility of cognitively demanding or interaction-heavy explanation formats. In contrast, deliberative domains such as healthcare or finance may allow for more detailed explanations that support calibrated trust and regulatory documentation. Consequently, insights into highly specialized applications may remain limited. Future research should explore how domain-specific constraints influence XAI method design and evaluation. Ideally, this research would integrate findings from application-driven studies into the broader landscape of user-centric XAI.

Our synthesis relies on the construct clustering proposed by Rong et al. (2023), which groups user-centric outcomes into broader categories such as trust, understanding, usability,

and human-AI performance. We adopted this structure to enable systematic aggregation and comparison across studies that employ highly heterogeneous evaluation measures. Without such clustering, synthesizing results across different operationalizations of user-centered outcomes would substantially limit the ability to identify general patterns. However, this aggregation also introduces limitations. Grouping diverse constructs abstracts methodological differences in how individual studies operationalize and measure outcomes. For instance, constructs grouped under the same category (e.g., trust) may represent conceptually distinct dimensions, such as self-reported perceptions or behavior-based indicators. Although we partially addressed this issue by distinguishing between implicit and self-reported trust where possible, further fine-grained differences may remain obscured by the clustering process. More broadly, relying on a single construct taxonomy may propagate its conceptual assumptions into the analysis. Future reviews could explore alternative construct mappings or more fine-grained categorizations to further examine how explanation effects vary across specific evaluation measures.

Beyond the uneven distribution of user groups and domains, the overall imbalance of the empirical literature also limits the generalizability of our synthesized guidance. Our recommendations are necessarily grounded on the patterns observable within the existing body of user studies, which disproportionately focus on lay users, certain explanation types (e.g., feature importance, counterfactuals), and specific application contexts. As a result, the strength of the derived guidance varies across contexts and XAI properties. While our synthesis identifies recurring trends in literature, these should not be interpreted as universally applicable across all domains, user groups, or emerging XAI methods. Future research should aim to broaden the empirical evidence base to validate and refine this guidance under more diverse conditions.

Overall, our review provides a solid basis for both theoretical refinement and practical guidance in user-centric XAI. However, more user studies are necessary to expand and diversify the evidence base.

7 Conclusion

The presented systematic literature review synthesizes empirical findings from 54 user studies on XAI. It emphasizes the importance of aligning XAI properties with user characteristics, task demands, and domain contexts. Our findings confirm that there is no universally optimal XAI method and highlight the importance of tailored approaches. Our analysis helps validate and refine existing theoretical guidance for selecting XAI methods. It emphasizes the interplay between user expertise and cognitive load in explanation effectiveness, as well as the importance of fostering calibrated trust rather than simply increasing it.

Our literature review also reveals significant gaps in the current XAI research landscape. There is limited empirical evaluation of several relevant XAI properties and sparse coverage of specialized domains. Addressing these gaps is essential for future research, which should aim to diversify empirical investigations and enrich the theoretical understanding of XAI effectiveness in various contexts. Additionally, our literature review offers practical guidance for system designers and practitioners seeking to implement tailored XAI methods in their applications. By linking XAI properties to specific user needs and task characteristics,

our findings help practitioners select suitable XAI methods that enhance user trust, understanding, usability, and human-AI performance.

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Declarations

Conflict of interest The authors declare no conflict of interest.

Ethical approval This article is a review of previously published studies and does not involve any new studies with human participants or animals performed by the authors.

Consent for publication Not applicable.

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